
SEIF EDDINE JOUL

Mechatronics Engineer

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SUMMARY

Multidisciplinary engineer proficient in instrumentation, PLC logic development, and diagnostic troubleshooting, strengthened via hands-on lab experimentation and advanced projects. Highly Skilled in integrated systems design as well as core electrical & mechanical systems commissioning. Equally confident planning and coordinating projects with peers, and applying statistical and analysis tools to drive technical decisions. With or without pressure, I stay organized and level-minded. I am enamored with innovation, to me engineering is all about novel and efficient problem-solving and optimizing existing solutions. Passionate about getting my hands dirty to turn theory and ideas into reality; aiming for the beauty of reliable and maintainable system design. I am flexible, ready to relocate, and thrive in both fieldwork and office-based environments. My reputation? Active team-player and reliable solo achiever.

EXPERIENCE

Electric/Electronic Technician - *Independent, Morocco*

Apr 2025 - PRESENT

- On-site technician in renovation and construction environment, responsible for lighting and power distribution; and ensuring preset schematics, color coding and safety standards are met during installation.
- Specialized in installation of CCTV and DVR surveillance systems, handling power supply and data transfer and network integration, in commercial and residential settings.

Assistant Engineer - *Hitronik, Malaysia*

Aug 2023 - Oct 2025

- Personally handled restoration, sensor recalibration and maintenance of university lab machinery and equipment, such as CNC lathes, as part of a maintenance team for the university.
- Co-led the development of multiple solar powered pump/valve automated systems using repurposed mechanical and electrical components, specialized in control/power circuitry design and on-site system installation.

Deputy President- *USM International Student Society, Malaysia*

Aug 2023 - Oct 2025

- Introduced a collaborative & results-driven approach during my tenure, creating new partnerships with adjacent campus societies and departments, significantly boosting our presence & reputation.
- Worked directly with our director to lead my team to plan and execute several large-scale multicultural initiatives and events that consistently advancing the society's vision and purpose, successfully achieving our annual objectives each time.
- Significantly increased our organizations outreach and visibility via increased social media presence, targeted marketing methods and focusing on a more personal approach regarding our initiatives; confirmed a x3 increase in social media monthly engagement and event turnout.

EDUCATION

Bachelor in Mechatronics Engineering - *Universiti Sains Malaysia, Malaysia* Feb 2020 - Nov 2024

Graduate with an Honours from a university ranked #128 globally (QS). Outcome-Based Education program accredited under the international Washington Accord. Focused on Automation and Instrumentation, with electives in Computer Vision and Quality, active athlete and volunteer on campus.

High school Diploma - *Al Nahda National School, UAE*

Jan 2016 - Jun 2019

SKILLS & CERTIFICATES

Skills: SolidWorks, PCB Design, Embedded Systems, PLC Programming (Ladder Logic, IEC 61131-3), Sensors & Instrumentation, OpenCV, Python, C/C++, Power BI, Minitab, Six Sigma (DMAIC), FMEA, Statistical Process Control (SPC), Root Cause Analysis, Agile (Kanban), Technical Documentation, Fault Isolation & Diagnostics, System Assembly & Wiring, Microsoft 365

Certificates in: PLC Sequencer Logic & Industrial Automation, Electrical Systems, Panel Boards, Frequency Drives & Schematics, Copper & Fiber Optic Communication Systems, Advanced Primavera P6, SAP ERP Essentials, Agile Project Management & Kanban, IELTS(C1 - 7.0), ICDL

PROJECTS

Drone Delivery Optimization & Reliability Modeling – *Data Analysis /Reliability Engineering*

Conducted drone delivery simulations using Minitab and one-way ANOVA across 5 payload platform configurations with 30+ simulation runs each. Applied regression modeling, stress analysis, and Weibull prediction techniques to evaluate multiple metrics, identifying a 2× increase in delivery time from platform size 2 to 10 while achieving >90% statistical confidence in performance behavior predictions

LiDAR-Camera Sensor Fusion System – *Computer Vision/Sensor Fusion*

Developed a MATLAB-based LiDAR-camera fusion system using extrinsic calibration, projective transformation, and YOLO object detection. Achieved sub-pixel alignment (<0.03 px reprojection error) and improved obstacle detection and depth estimation accuracy by ~35% compared to individual camera sensor.

4-DOF Robotic Arm (Simulation /Fabrication /Control) – *Robotics /Control Systems*

Designed and fabricated a servo-actuated robotic arm integrating forward/inverse kinematics, Cartesian motion control, and custom GUI interface using Arduino, LabVIEW, and SolidWorks. Achieved end effector positioning accuracy within ± 3 mm and response latency below 100 ms.

Vacuum Pick-and-Place Handling Station – *Industrial Automation /Pneumatics*

Designed and commissioned a PLC-controlled pneumatic pick-and-place station using ladder logic AND structured text on OMRON CX-ONE. Achieved repeatable transfer cycles under 5 seconds with 0.5 mm positioning accuracy while improving system reliability by ~70%.

International Food and Culture Fiesta – *Leadership/Event Management*

Led the organization of a multicultural university event hosting 500+ attendees with a managed budget exceeding RM 20,000. Coordinated logistics, media, sponsorships, and team operations to deliver one of the university's flagship student events.